Measuring Earth's Magnetic Field with a Smart Device

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Aim:

Measure the strength and inclination of Earth's magnetic field at your school using a magnetometer app on a smart device, then compare your results to the current World Magnetic Model.

Materials:

- Mobile phone/ tablet with the phyphox app installed
- Protractor
- String/ twine (approx. 60 cm)
- Weight e.g., small shell/ pebble

Activity 1: Predicting the value of magnetic inclination at your location

Magnetic inclination is the angle that Earth's magnetic field lines make with Earth's surface (GFZ, 2025).

Inclination is ± 90° at the poles and 0° at the equator.

A simple diagram showing how magnetic inclination changes with latitude is shown in **Figure 1.**

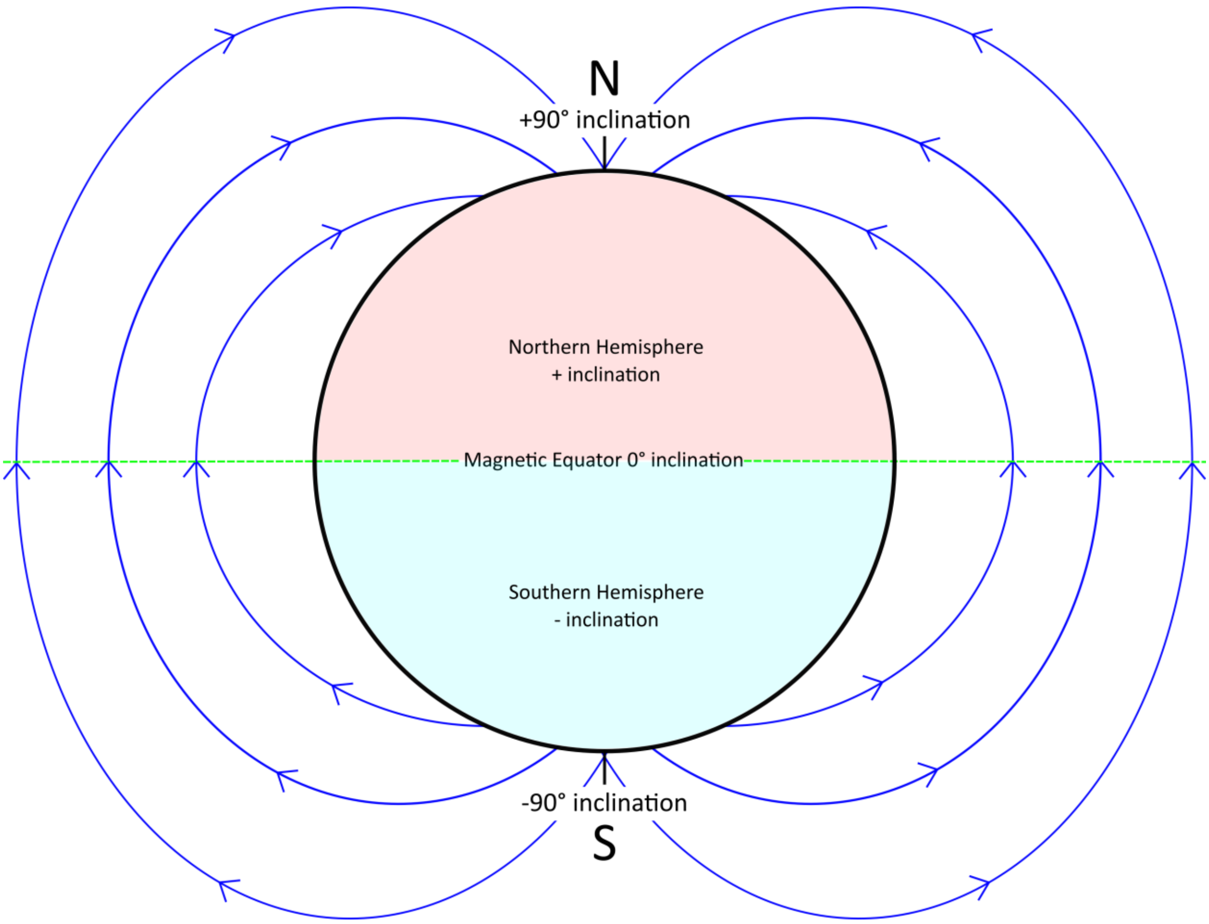

Figure 1: Simple diagram of Earth's magnetic inclination. Magnetic field lines and their direction are shown in blue.

1. What do you predict that the value of magnetic inclination will be at your location?

Activity 2: Measuring magnetic field strength and inclination using phyphox

Step 1: Assembling a Plumb Bob to Measure Magnetic Inclination

Demonstration video: Measuring Earth's Magnetic Field with a Smart Device Step 1: Assembling a Plumb Bob

We will now construct a simple plumb bob that we can use to measure magnetic inclination on a smart device.

A **plumb bob** is simply a weight attached to a string. It is used as a vertical reference line to help measure angles.

1. Tie your weight e.g., a small pebble/ shell to one end of your string.
2. Tie the other end of your string around the short edge of your device so that there is approximately 20 cm of length hanging down.
3. Hold a protractor against the long edge of your device, with the plumb bob string against the outside of it.

Your set up should look like **Figure 2.**

We will use the angle that the string makes against the protractor to measure magnetic inclination.

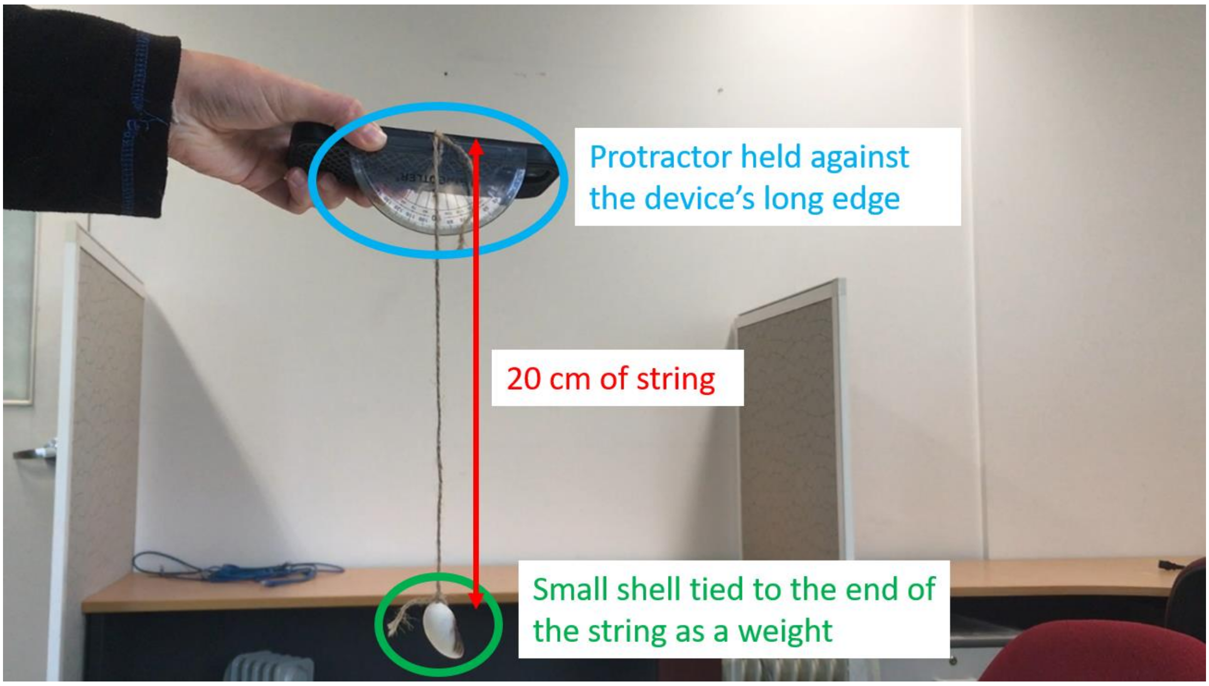

Figure 2: Labelled plumb bob set up.

<u>Step 2: Measuring Magnetic Field Strength & Inclination</u>

Demonstration video: [Measuring Earth's Magnetic Field with a Smart Device Step 2: Measuring Field Strength & Inclination]

We are now going to measure a value for Earth's magnetic field strength and an angle of magnetic inclination in your classroom.

Note: You will need to have the phyphox app downloaded on your device for this step. See the document *'Downloading and using the phyphox magnetometer'* for guidance.

1. Hold your device (including the plumb bob and protractor) horizontally out in front of you.
2. Start recording with the phyphox magnetometer in **Graph** mode.
3. Bring the value of **Bx to 0** by rotating your body whilst keeping your device flat (**Figure 3**).
4. When you have found the position where Bx is 0, keep your body still and bring **By to 0** by tilting your device vertically up/ down (**Figure 4**).
5. Hold your device in the position where Bx and By are both 0. Note what value **Bz** stabilises at – this is the magnetic field strength at your location. Record this in **Table 1**.
6. Also note what **angle** the plumb bob string makes against the protractor in this position – this is the angle of magnetic inclination at your location. Record this in **Table 1**.

Note: you may find it easier to get a partner/ member of your group to read off the angle of inclination whilst you hold everything still.

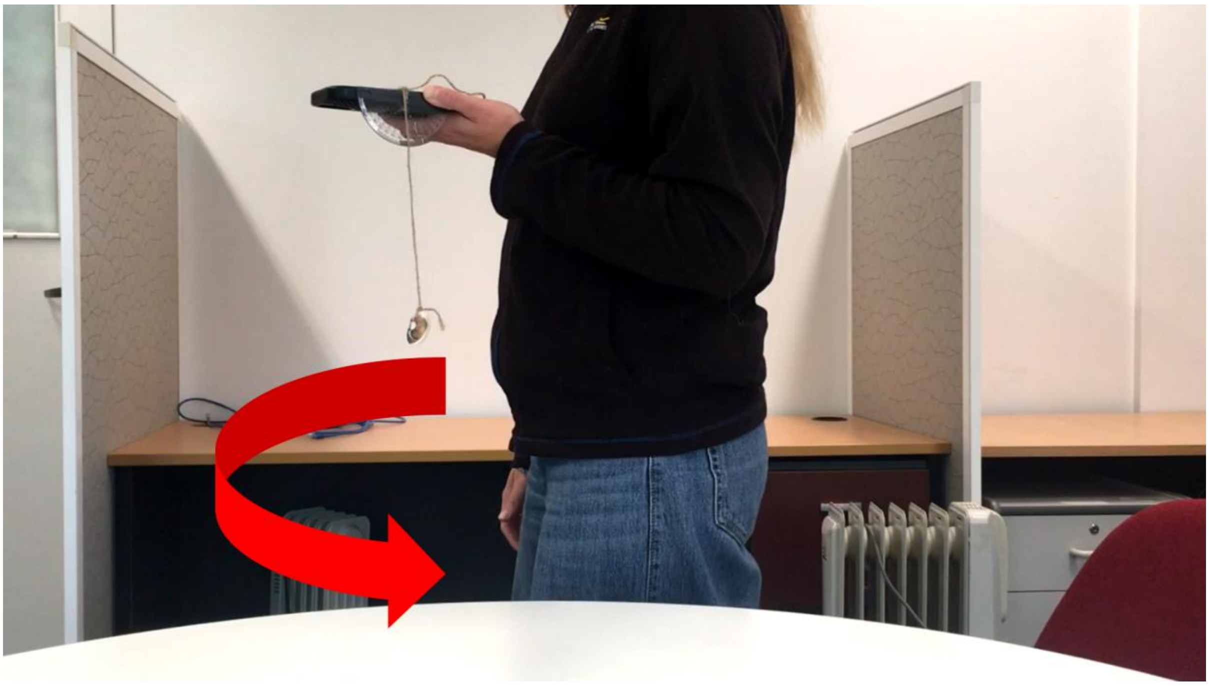

Figure 3: Bringing Bx to 0 by holding the device flat and turning your body.

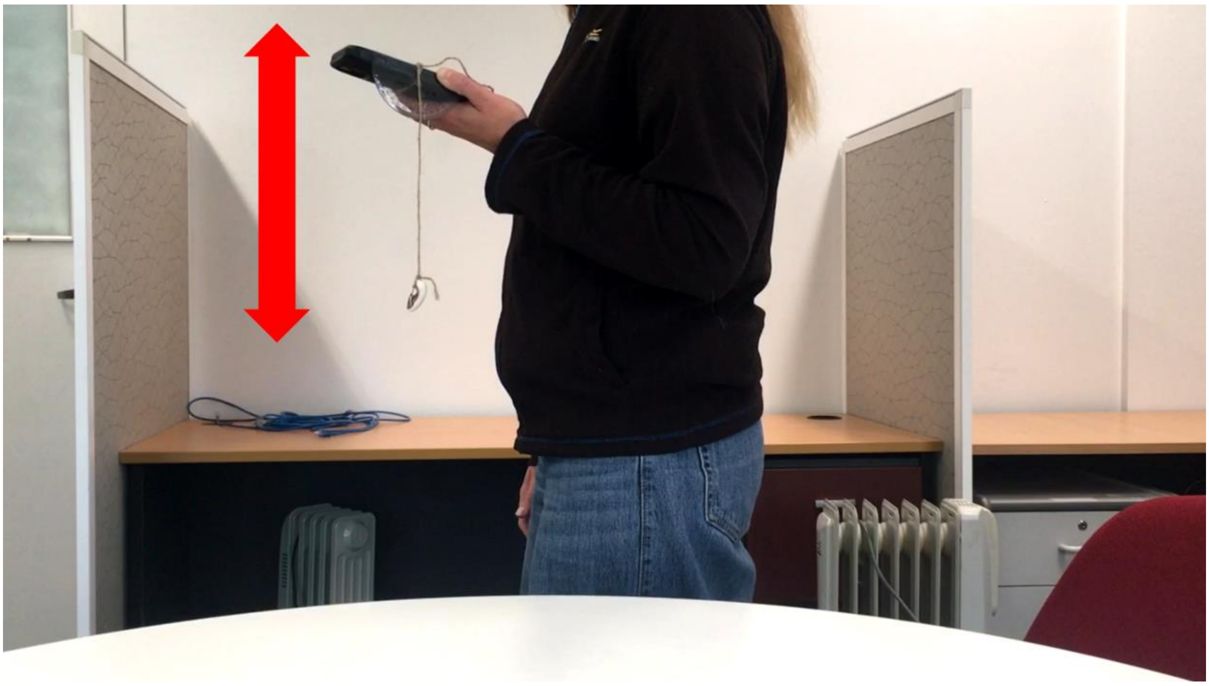

Figure 4: Bringing By to 0 by keeping your body still and tilting the device up/ down.

Note: You will notice that there are 2 opposite orientations you can face where Bx and By are both 0. You can measure the magnetic field from either direction.

If your device is tilted up, the magnetic inclination angle will be negative. If it is tilted down, the angle will be positive.

Extension question:

Why do you think we bring Bx and By to 0 to measure Earth's magnetic field?

Your teacher will now take you outside onto your school field/ oval to repeat your measurements of magnetic field strength and inclination.

Record the values you collect outside in **Table 1.**

Activity 3: Comparing your results to the WMM

1. Is there a difference between the values of field strength and magnetic inclination you collected inside and outside?

2. If there is a difference, what do you think caused this?

Figure 5 and **Figure 6** are maps of Main Field Total Intensity (F) and Main Field Inclination (I) from the **World Magnetic Model** (WMM).

The maps are also available on the WMM webpage here under the 'Maps' tab.

What is the WMM?

The World Magnetic Model (WMM) is the standard model of the Earth's geomagnetic field. It is updated every 5 years to reflect any changes in the field. The model is developed by the British Geological Survey and the NCEI (NCEI; BGS, 2024).

Find your location on the WMM maps and read off the value for magnetic field intensity (aka strength) and magnetic inclination. Record these in **Table 1**.

Note: on the WMM maps, magnetic field intensity is given in nanoteslas (nT) whilst the phyphox app reports values in microtesla (μT).

To convert from nT to μT, divide the value by 1000. E.g., 50,000 nT = 50 μT.

1. How well do the values you collected compare to the values from the WMM?

2. If they aren't a perfect match, can you think of any sources of error or uncertainties in your experimental set up?

Table 1: Magnetic field strength and magnetic inclination values.

	Magnetic Field Strength (μT)	**Magnetic Inclination (°)**
Inside		
Outside		
WMM values		

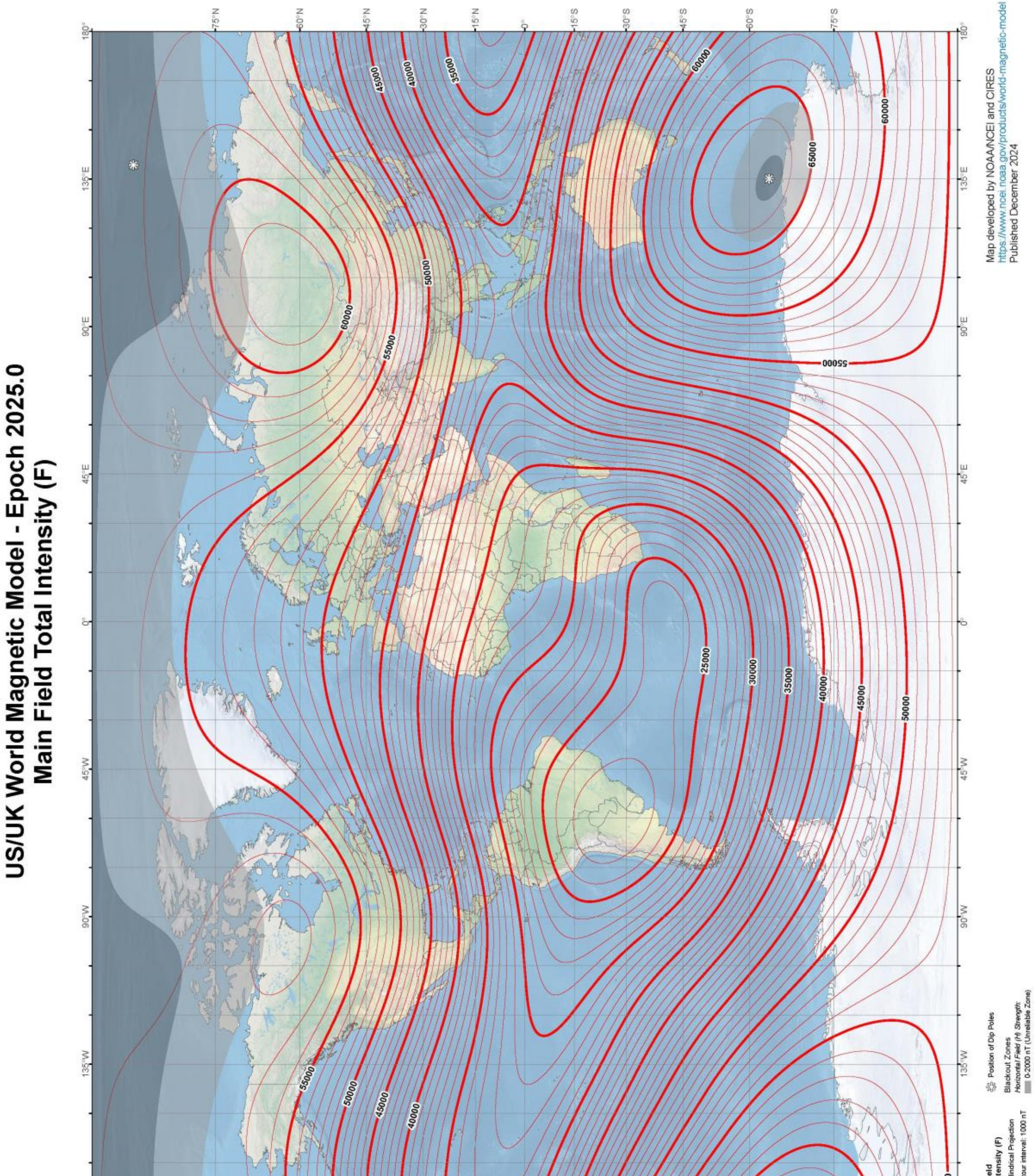

Figure 5: WMM map of magnetic field strength (NCEI; BGS, 2024).

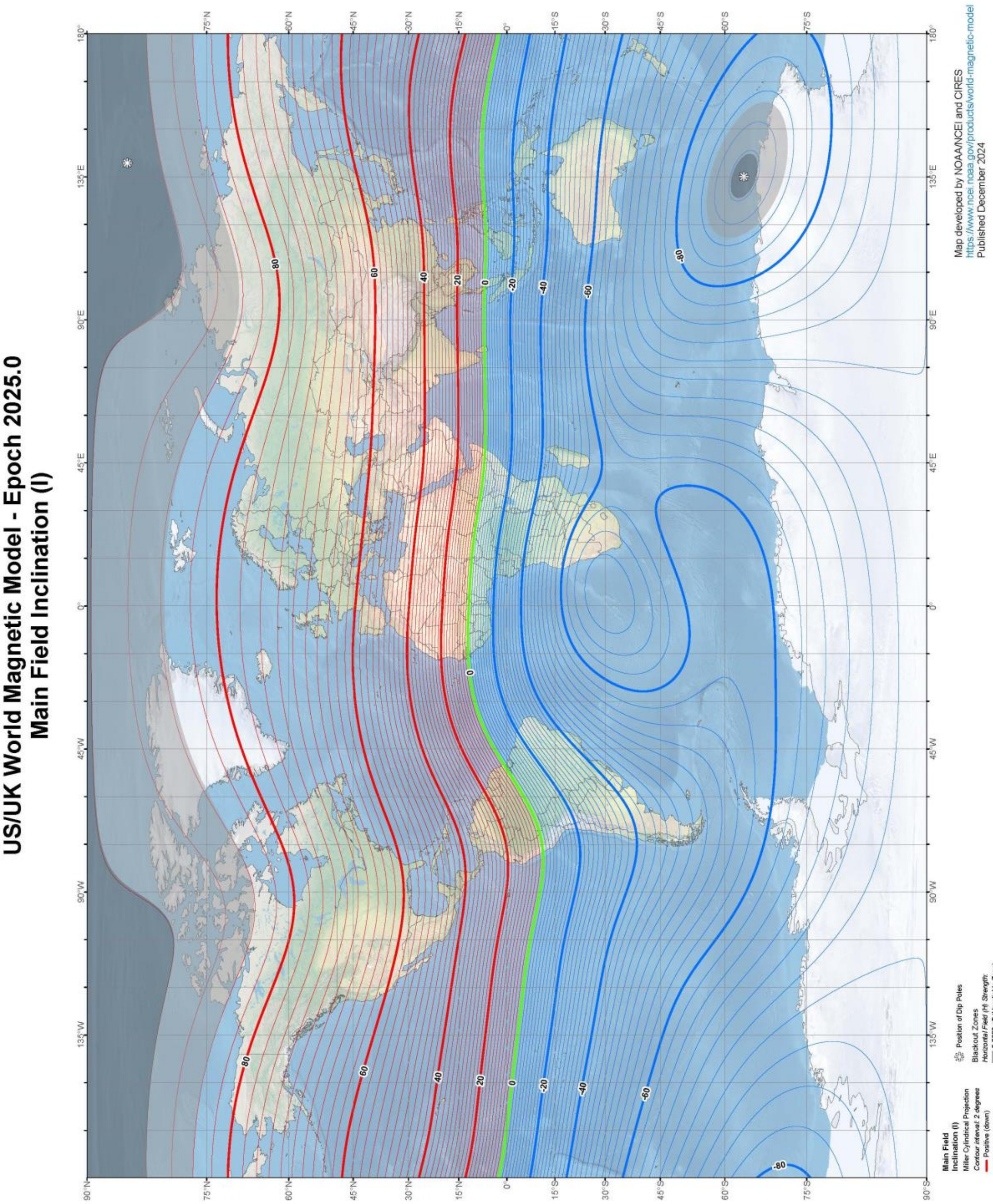

Figure 6: WMM map of magnetic field inclination (NCEI; BGS, 2024).

References:

GFZ German Research Centre for Geosciences (2025). *Frequently asked questions regarding geomagnetism*. Available at: https://www.gfz.de/en/section/geomagnetism/data-products-services/frequently-asked-questions-regarding-geomagnetism#c14403

NOAA NCEI Geomagnetic Modeling Team; British Geological Survey. 2024: World Magnetic Model 2025. NOAA National Centers for Environmental Information. https://doi.org/10.25921/aqfd-sd83